DDR Installation Procedure (260iB & DDR iB)

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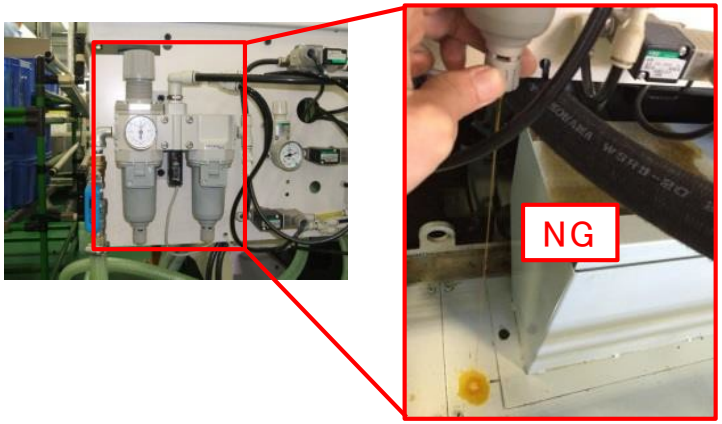
1. Airsupplied to DDR

1 ・Compressed air environment (air pressure, air quality)

The air pressure supplied to DDR is 0.35～0.5MPa.
Use clean air with less moisture(water), oil and dirt.

～Compressed air supply～

0.35 to 0.5MPa (gauge pressure) (0.5MPa is recommended)
0.15m³/min (150L/min) (at atmospheric pressure)
Solid material 5 μm or less (nominal)
Moisture Dew point under atmospheric pressure −17° C or below
Oil content concentration on the primary side 1 mg/m³ or below0.5 MPa.



※ If the air contains a lot of oil or moisture, or if the air pressure is out of the specification range,the unclamping operation may not be performed normally and operation delay may occur, that leads to the clamping mechanism parts wear during rotation operation.(Right Fig.)

※ If the air contains a lot of moisture, the O-ring of the spool in the solenoid valve will be poor lubrication, that leads to wear out.
And rust occurs in the air flow path. (Right Fig.)
As a result, the unclamping operation is not performed normally, and the clamping mechanism parts wear during rotation operation.

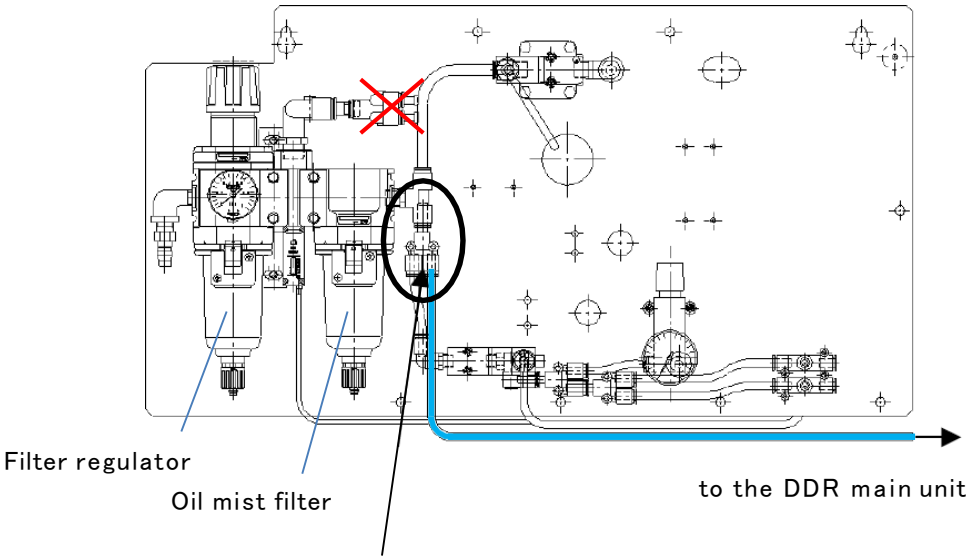
Lubrication defect of spool (sliding part)
⇒Wear of O-ring

Rust in the air flow path

2. Air connection to DDR

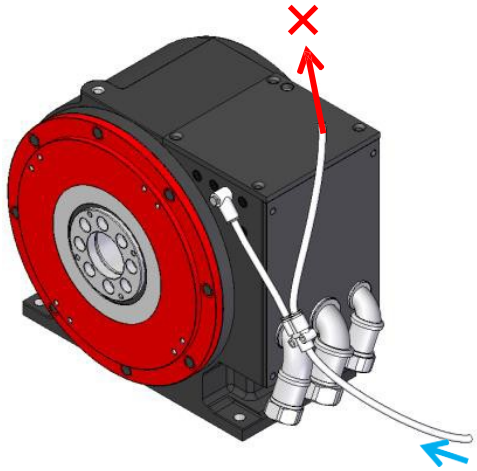
2 ・Air connection

Be sure to connect the air tubes at the position shown in the figure.
Otherwise, oil, liquid, and foreign matters may get in the DDR, causing a failure in the DDR solenoid valve, clamp mechanism, or another part.



Replace the plug reducer and Y-joint (appended goods).
Connect the original air tube to one of them, and the air tube for the DDR to the other.

Do not branch the air that is to be supplied to DDR.
(Only supply of internal pressure for Tail support made by FANUC can be branched)



※If air bifurcated on the way is used for air blow etc., the air pressure to be supplied to DDR decreases causing a decrease in clamping torque, and clamp/unclamp operation will failure because of insufficient flow rate, that may cause wear of clamp mechanism parts.

3. Parameter setting

3 - 1 ・Keep relay setting

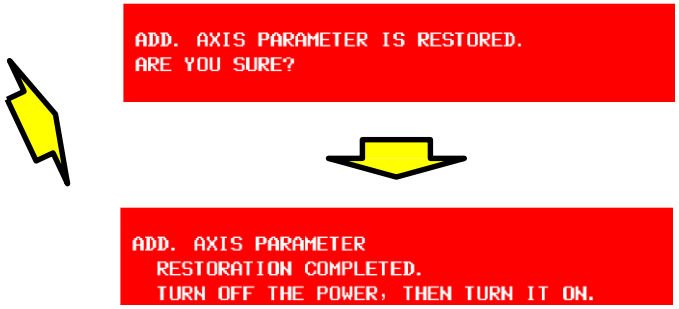
(1) When using DDR, the keep relay settings below are required

K32#1=1 (to enable 4-axis DDR control)
K15#4=0 (to disable DDR260i control)
K02#1=0 (not to shut off servo excitation for fourth-axis clamping)
K03#2=1 (to use fourth-axis clamping and unclamping)

※When using DDR-T or DDR + other company's support table <with brake>, the keep relay setting below is required too

K32#3=1 ※Item related to timer setting (on next page)

(2) By performing the above setting and re-setting the additional axis parameters, the parameters of DDR are read.



3. Parameter setting

3 - 2 · Timer setting

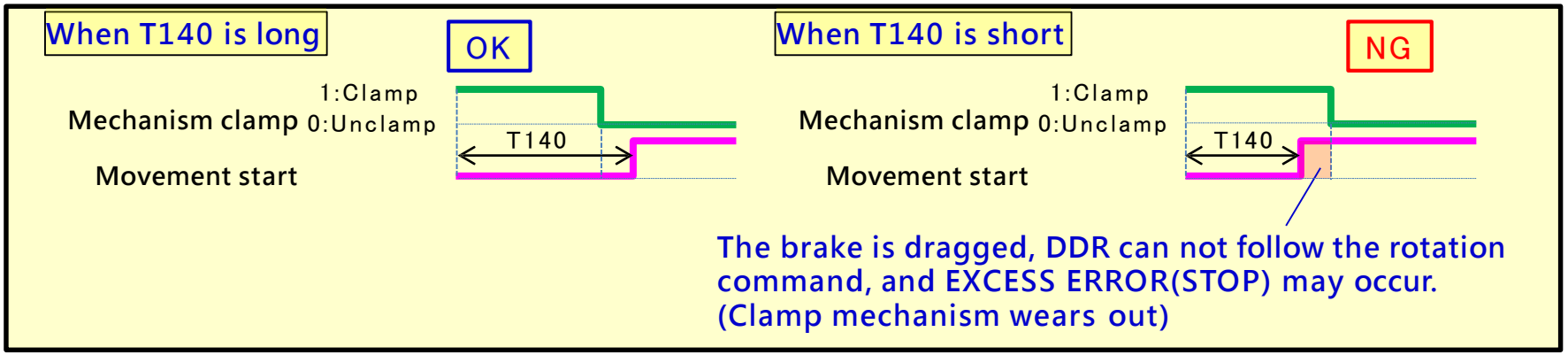
Timer	K32#3=0 •DDR •DDR + Support table made of the other companies(without brake)	K32#3=1 •DDR-T（DDR + FANUC tail support） •DDR + Support table made of the other companies(with brake)
T140	184 and above	376 and above
T162	168 and above	200 and above
T172	168 and above	200 and above

（ ms）

T140 : Interlock time (in ms) since B-axis unclamping

T162 : Time (in ms) since clamping to torque limit change and interlock

T172 : Time (in ms) from clamping to clamp check start

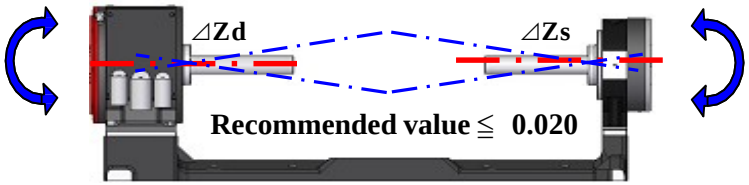
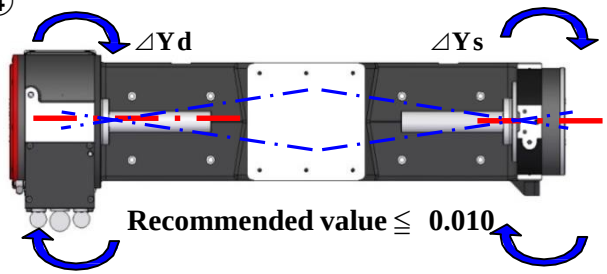
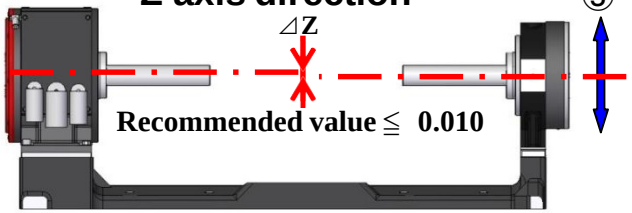
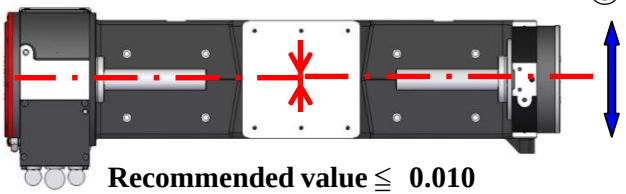
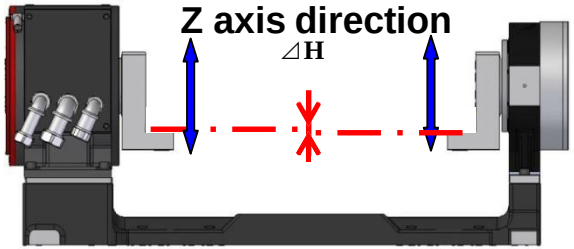


For ROBODRILL shipped after March 2011, the timer value is automatically entered according to the keep relay setting described on the previous page;
But for previous ROBODRILL, you need to change T140 value manually.

4. Alignment of DDR and support table

When the DDR is misalignment with support table, the load depending on the angle is applied to the DDR.

* ・EXCESS ERROR、vibration may occur during rotation.
・ Life of bearing may be shortened.

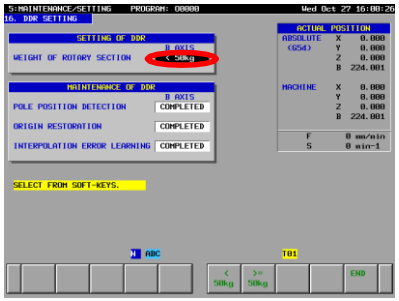
Axis slant adjustment	Around Y axis ① ②  	Around Z axis ④ ⑤ 
Axis center alignment	Z axis direction ③  	Y axis direction ⑥ 
Adjustment of bracket height		Z axis direction ⑦ 

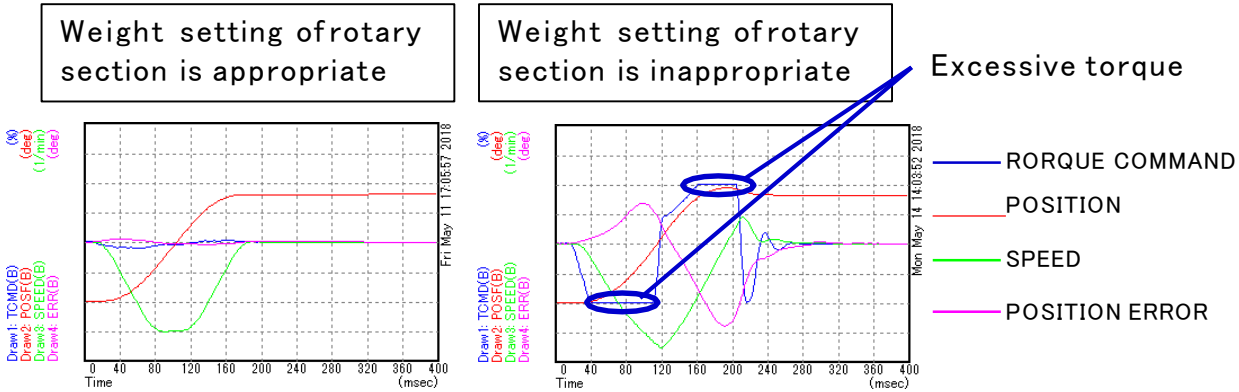
5. Parameter setting (Weight setting of rotary section)

5・Weight setting of rotary section

Main parameters depending on weight setting

- Rapid traverse rate (No.1420)
- Maximum allowable acceleration rate for rapid traverse (No.1671)
- Bell-shaped time constant for rapid traverse acceleration/deceleration (No.11242)
- Velocity gain





When select a setting which is lighter than the actual total weight of the jig and workpiece, the operating conditions become strict. The output torque of motor at acceleration/deceleration becomes maximum, and OVC alarm or overheat alarm occurs.
And, DDR can not follow the rotation command, that causes excess error (stop).

For DDR(DDR-T), it is necessary to set the weight of rotary section according to the weight of jig and workpiece.

※But due to configuration of jig and workpiece, even weight setting of rotary section is appropriate, oscillation of rotation or fluctuation of stoppage may occur, that cause EXCESS ERROR.

Adjustment of Velocity Gain & Resonance Elimination Filter is necessary
(as will hereinafter be described in detail)

6. Parameter setting (Servo)

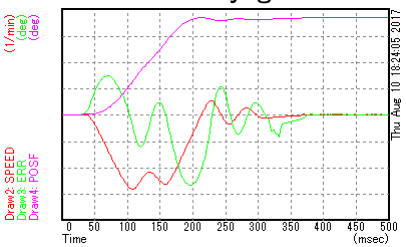
6 · Servo parameters (Velocity Gain, Resonance Elimination Filter) (1 / 3)

Servo parameters are important parameters determine the response of the servomotor. In order to improve responsiveness, it is necessary to set the value of velocity gain high.
However, excessive velocity gain may cause vibration due to mechanical resonance or the like. By using resonance elimination filter to avoid vibration, and adjust velocity gain as high as possible.
Depending on the structure of jig and workpiece, appropriate parameter setting values are different.

＊ Need to set for each set of jig and workpiece.

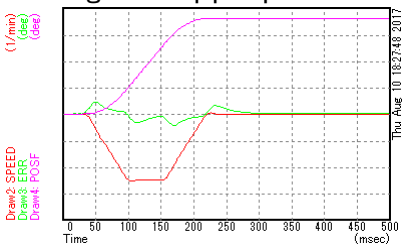
Adjusting the velocity gain

When velocity gain is small



Followability is bad
Position deviation is large
＊ Excess Error occurs

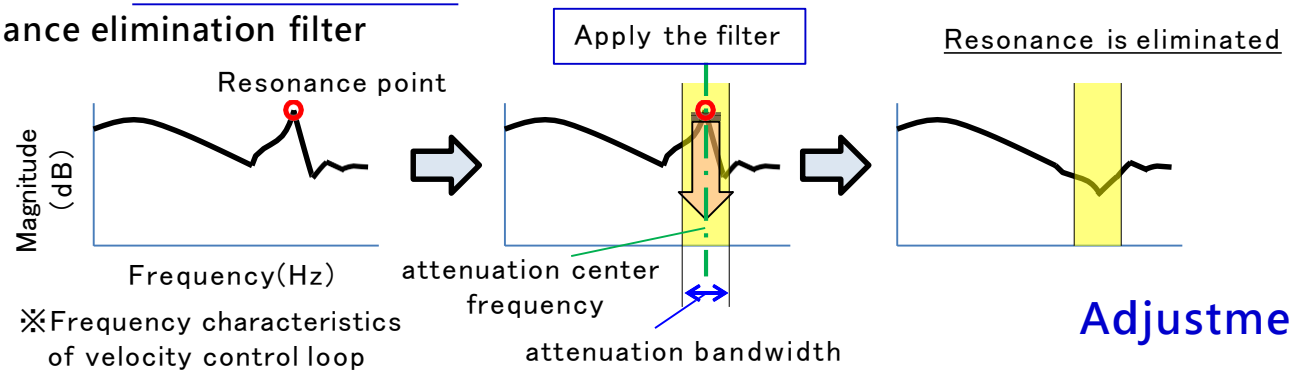
After setting the appropriate velocity gain



Followability is good
Position deviation is small

SPEED
POSITION
POSITION ERROR

Adjusting the resonance elimination filter



Adjustment method

6. Parameter setting (Servo)

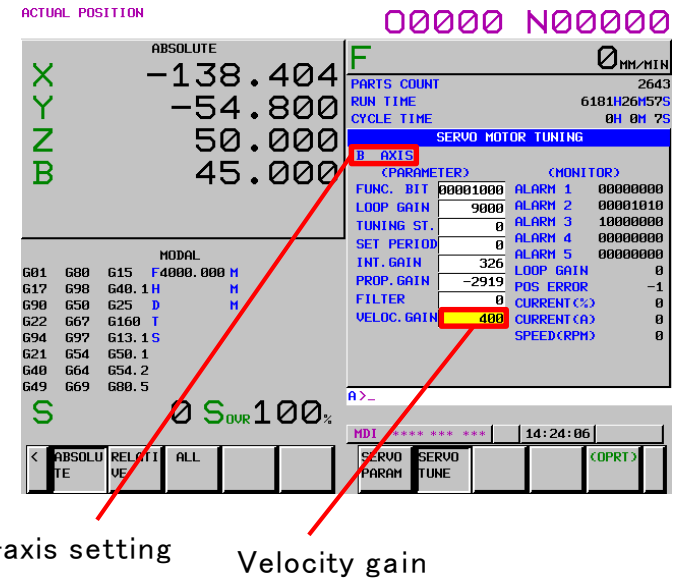
6.1・Servo parameter (Velocity Gain, Resonance Elimination Filter) (2 / 3)

①Adjusting the velocity gain

1. After setting weight of rotary section, change PMC keep relay K00#6=1, then restart power
2. Display the SERVO MOTOR TUNING screen by selecting [SYSTEM]⇒[+]⇒[SERVO PARAM] ⇒[SERVO TUNE] in the stated order.
3. Display the B-axis setting screen by selecting [PAGE ↓].
4. Increase the [Velocity gain] value in steps of 50%, and rotates the DDR.
5. Increasing the velocity gain gradually may cause the following oscillation phenomena.
   - The DDR main unit generates vibration noise.
   - The current value shows signs of disorder.
   - Positional deviation during stopping becomes large.
6. Use 70% to 80% of the velocity gain when one of the above oscillation phenomena occurs as the final velocity gain setting.
7. Perform rotation, clamping, unclamping, etc. to check that there is no problem with the operation.

※After finish the final velocity gain setting as above, if the following problems occur, please adjust resonance elimination filter

- DDR vibrates during rotation or stop
- Noise does not disappear
- Excess error (stop) occurs immediately after rotation



6. Parameter setting (Servo)

2 - 4 · Servo parameter (Velocity Gain, Resonance Elimination Filter) (3 / 3)

Parameters for the resonance elimination filter

- Filter 2~3 have been set as standard parameters
- New addition to filter 1 or filter 4 is possible

(※ However, value of filter 1 is automatically set when weight setting of rotary section is 150 kg)

	Attenuation center frequency [Hz]	Attenuation bandwidth [Hz]
Resonance elimination filter 1	No.2113	No.2177
Resonance elimination filter 2	No.2360 (= 1100)	No.2361 (= 600)
Resonance elimination filter 3	No.2363 (= 700)	No.2364 (= 200)
Resonance elimination filter 4	No.2366	No.2367

	Attenuation center frequency [Hz]	Attenuation Bandwidth [Hz]
Setting 1	200	100
Setting 2	250	100
Setting 3	300	100
Setting 4	350	100
Setting 5	400	150
Setting 6	450	150
Setting 7	500	200
Setting 8	550	200
Setting 9	600	200
Setting 10	650	200
Setting 11	700	200
Setting 12	750	200
Setting 13	800	200
Setting 14	850	200
Setting 15	900	200
Setting 16	950	200
Setting 17	1000	300
Setting 18	1050	300

②Adjusting the resonance elimination filter

1. Return the velocity gain to the value when oscillation occurs on「①Adjusting the velocity gain」step.
2. Input each value of the setting table on the right into the filter 4, then set the one which let the oscillation be most reduced when the axis is rotated.
3. Again, perform「①Adjusting the velocity gain」more one time, and set the velocity gain to 70～80% of the value when oscillation occurs.
   Confirm that vibration, abnormal noise, excessive error alarm does not occur.
   (If the problem still persists, please make additional setting to Filter 1, and adjust the velocity gain again.)

7- Set up List

No.	Item		・DDR ・DDR + Support table made of the other companies（without brake）	・DDR-T ・DDR + Support table made of the other companies（with brake）
1	Keep relay	K32#1	1	1
		K15#4	0	0
		K02#1	0	0
		K03#2	1	1
		K32#3	0	1
2	Timer	T140	184 and above	376 and above
		T162	168 and above	200 and above
		T172	168 and above	200 and above
3	Weight setting of rotary section		Set the weight of rotary section according to the weight of jig and workpiece	
4	Servo parameter		Adjust for no oscillation, noise, and fluctuation when stopped	
5	Alignment of DDR and support table（When using support table）		Center the axis correctly	
6	Compressed air (air pressure, air quality)		Air pressure is within specification range.（0.35～0.5MPa）	
			Use air that does not contain moisture and oil	
7	Air branching method		Do not branch the air to be supplied to DDR （Except for internal pressure of Tail support made by FANUC）	

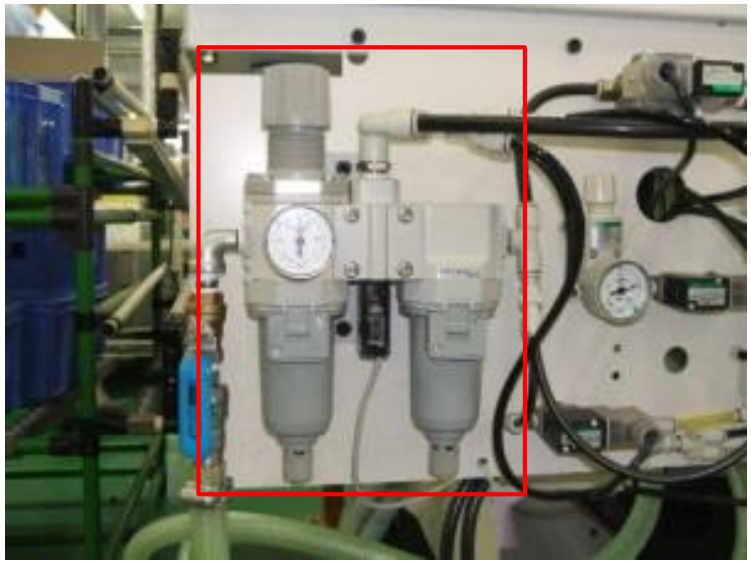
8- Periodic checking of DDR

No.	Periodic checking	Avoidable failures by check and pre-deal	Checking frequency
1	Checking air pressure	Machining failure due to low clamp torque or EXCESS ERROR (STOP) alarm occur	1 time／day
2	Visual checking ①Whether there is air leakage from the oil seal of the terminal box ②Whether the drain has been blocked with metal chips	①The internal pressure is lower ⇒Coolant may get into the DDR ②The internal pressure is higher ⇒Extrusion of O-Ring from terminal box ⇒lower pressure ⇒Coolant intrusion	1 time／day
3	Checking internal pressure	Insulation Degradation by coolant intrusion	1 time／month
4	Checking clamp torque	Machining defects	1 time／month
5	Checking filter regulator and oil mist filter	Clamp/unclamp operation failure（failure of valve assy, clamp unit）	1 time／month

8.1. Checking air pressure

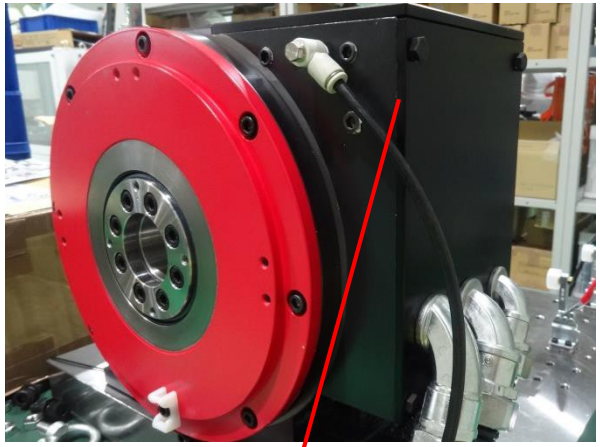
～Compressed air supply～

0.35 to 0.5MPa (gauge pressure) (0.5MPa isrecommended)
0.15m^3/min (150L/min) (at atmospheric pressure)
Solid material 5 μmor less(nominal)
Moisture Dew point under atmospheric pressure −17° C or below
Oil content concentration on the primary side 1 mg/m^3 or below0.5 MPa.

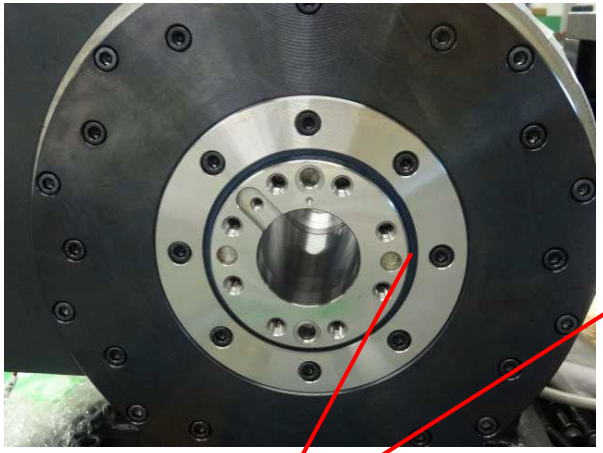


- The air pressure supplied to DDR is0.35～0.5MPa.
- Use clean air with less moisture(water), oil and dirt.

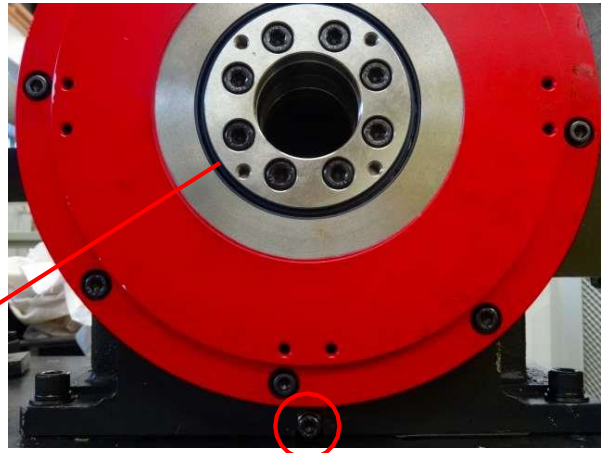
8.2- Visual checking



Terminal Box



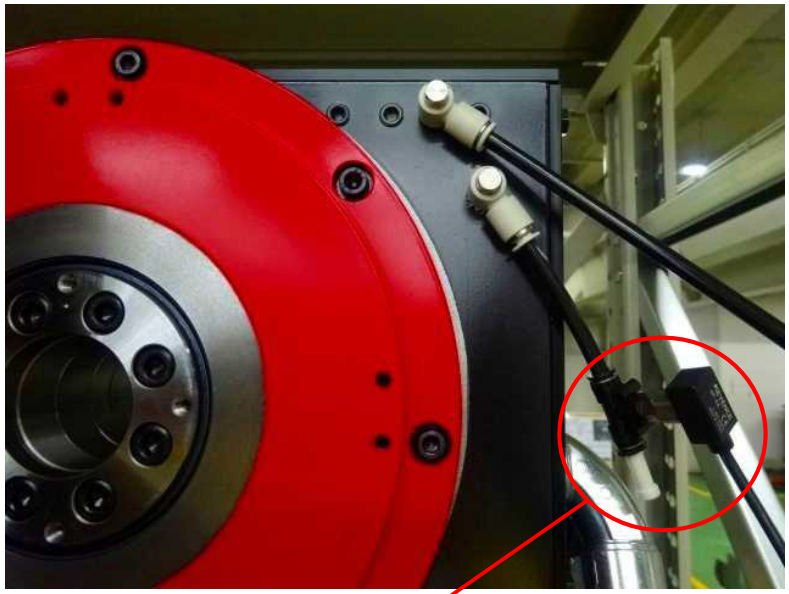
Oilseal



Drain

- Make sure there is no air leakage from the oil seal of the terminal box
- Make sure the drain is not blocked with metal chips

8. 3 Checking internal pressure



Internal pressure (At base pressure is 0.5 MPa)	Decision
10 kPa or higher	The internal pressure is higher than the rated value. Check to see if the drain hole have been blocked with metal chips and whether the speed controller is abnormal.
2 to 10 kPa	Good
Below 2 kPa	The internal pressure is not sufficiently maintained. Check the sealing parts and adjust the air purge internal pressure, using the air purge pressure adjustment bolt.

- Measure the air purge pressure by using a pressure gauge

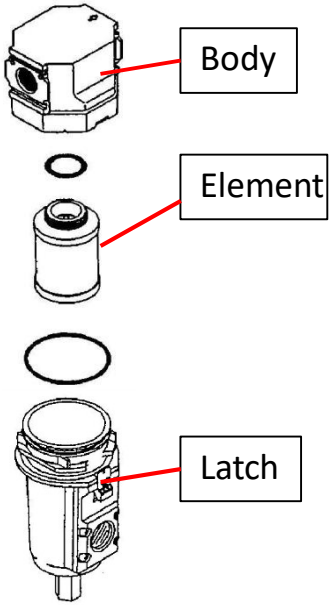
8. 4-Checking clamp torque (Brake test)

➢ At the air pressure supplied is 0.5 MPa, Rotate a small angle (0.05°) and check the load meter

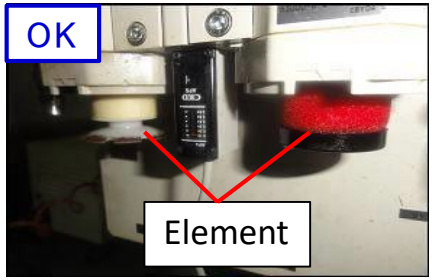
(1)Change the settings of the keep relay and data table as listed below.
Keep relay K01#5 = 1
Data table D154 = 255
(2)Issue M71 to start a clamping operation.
(3)Select the manual mode, and press the <4th> button.
(4)Turn the handle toward the "+" side to set it to 0.050° from the current position.
(5)The brake disc is normal if the load meter reads 500% or higher and if alarm SV0436, "SOFTTHERMAL (OVC)", or EX1089, "DDR OVC (DDR-4) alarm", occurs within 10 seconds.
(6) Return keep relay K01#5=0 and data table D154=25 to their respective original values.
(7) Turn the power off and on again.

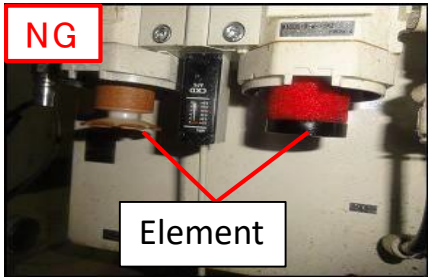
Value of load meter	Decision
500% or higher and OVC alarm occurs within 10 seconds	Clamp torque is good
Below 500%	Clamp torque is NG There is a problem with the clamp mechanism parts

8.5 - Checking filter regulator and oil mist filter



➢ Check the state of element





Element replacement is necessary when dirt is progressing.
When used for a long time in this state, the air which could not completely remove moisture and oil will flow to the DDR, that may make clamp / unclamp operation delay or failure.
⇒ It is necessary to replace valve assy and O-Rings of clamping mechanism.

9. Maintenance List

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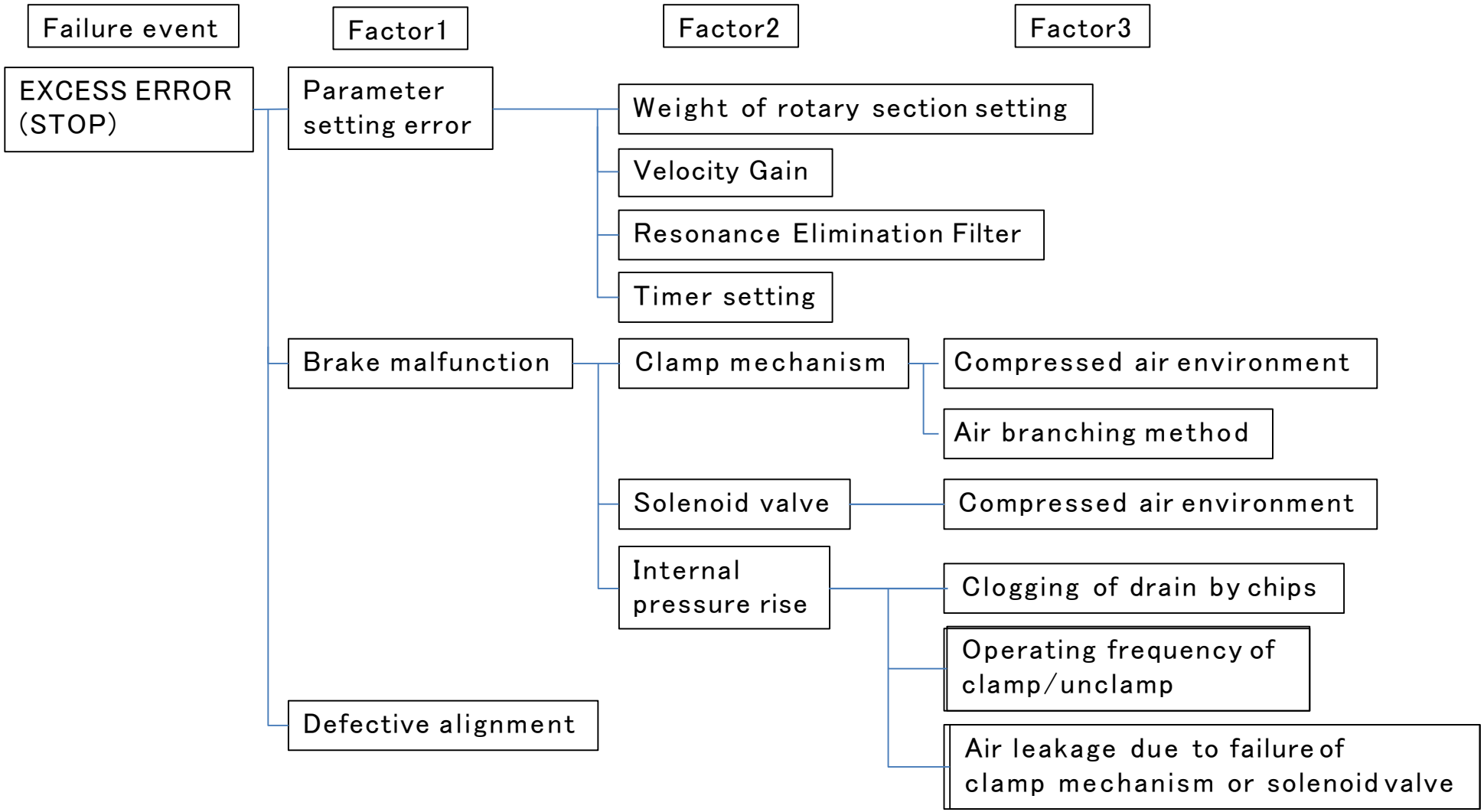
No.	Periodic inspection	Detail	Inspection frequency
1	Air pressure	Whether the pressure is in the range of 0.35～0.5MPa	1 time / day
2	Visual inspection	Whether there is air leakage from the oil seal or the terminal box	1 time / day
		Whether the drain has been blocked with metal chips	1 time / day
3	Internal pressure	Whether the internal pressure is in the tange of 2～10kPa	1 time / month
4	Clamp torque	Whether the clamp torque is appropriate	1 time / month
5	Filter regulator and oil mist filter	Whether there is water or oil accumulating in filter regulator and oil fiter	1 time / month
		Whether the mantle (element) is dirty	1 time / month

10. Trouble shooting chart of B axis EXCESS ERROR(STOP)